FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2024

TEST DATE: 24-03-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

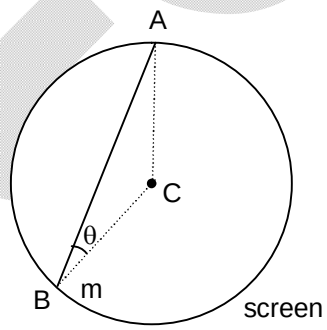
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The displacement x of a particle moving along a straight line at time t is given by $x = -\frac{1}{2}(t-1)^2 + 2(2-t)^2$. The velocity of particle is zero after time……..

(A) $\frac{15}{3}$ sec (B) 5 sec

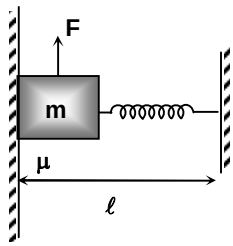
(C) 17 sec (D) $\frac{7}{3}$ sec

2. AB is a string whose one end A is fixed at the highest point of a ring placed in vertical plane. C is the centre of the ring. At other end B of the string, is attached a bead of mass m, which can slide along the ring. If the bead is in equilibrium at B and string is taut then the tension in the string is : (given m = 0.2 kg, θ = 30°, g = 10 m/s^2)



(A) 4√3 N
(B) 6√3 N
(C) 2√3 N
(D) None of these

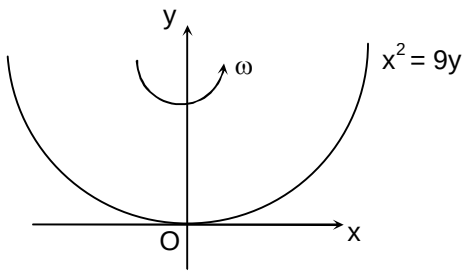
3. A block of mass m is pressed against a vertical surface by a spring of unstretched length 2ℓ and spring constant K. If the coefficient of friction between the block and the surface is μ. The minimum value of force F applied on the block in vertically upward direction so that block just remain equilibrium. (given ℓ = 1 m, μ = 0.1, m = 5.4 kg, g = 10 m/s^2, K = 540 N/m



(A) 2 N (B) 0

(C) 3 N (D) 4 N

4. Vertical section of a bowl shown in figure can rotate about axis OY. The section is parabolic in shape and represented by the equation $X^2 = 9Y$. Then find angular velocity of the bowl for which a sphere of small size placed on its smooth inner surface does not slide:



(A) $\omega = \sqrt{\frac{9}{2g}}$ (B) $\omega = \sqrt{\frac{2g}{9}}$

(C) $\omega = \frac{2g}{9}$ (D) $\omega = \frac{9}{2g}$

5. A stone is tied to a string of length 2m is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed 10m/s. The magnitude of the change in its centripetal acceleration as it reaches a position where the string is horizontal, is :

(A) $6\sqrt{5}$ m/s^2 (B) $3\sqrt{5}$ m/s^2

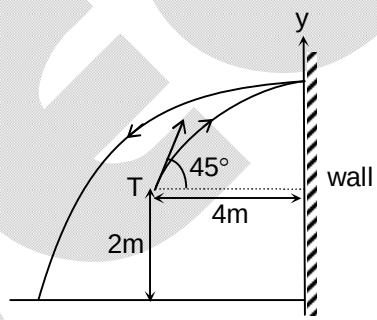
(C) $4\sqrt{5}$ m/s^2 (D) $10\sqrt{34}$ m/s^2

6. A ball A moving with kinetic energy E, makes a head on elastic collision with a stationary ball with mass 2n times that of A. The maximum potential energy due to deformation stored in the system during the collision is :

(A) $\frac{2nE}{(2n+1)}$ (B) $\frac{(n+1)E}{n}$

(C) $\frac{(n-1)E}{n}$ (D) $\frac{E}{n}$

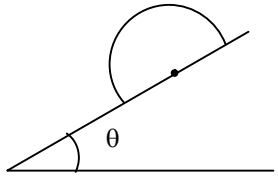
7. A perfectly elastic ball is thrown towards a smooth wall and bounces back over the head of the thrower (T), as shown in the figure. When it leaves the thrower's hand, the ball is 2m above the ground, 4m away from the wall and has initial velocity of $10\sqrt{2}$ m/s at an angle of 45° to horizontal. The ball will hit the ground at an approximate horizontal distance of :



(A) 18m from thrower (B) 12.2m from thrower

(C) 10.6m from thrower (D) 13.8m from thrower

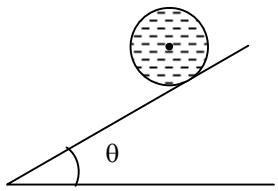
8. An uniform hemi-hollow sphere is placed with flat surface on rough inclined plane as shown in figure. If friction is large for no sliding, then the minimum angle θ at which toppling occur is :



(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) 45°

(C) $\tan^{-1}(2)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$

9. A thin hollow sphere of mass m is completely filled with non-viscous liquid of mass m . When the sphere rolls without sliding on inclined plane. Find the linear speed of the sphere when it covers distance ℓ on the incline plane



(A) $v=\sqrt{2g\ell\sin\theta}$ (B) $v=\sqrt{\frac{5}{2}g\ell\sin\theta}$

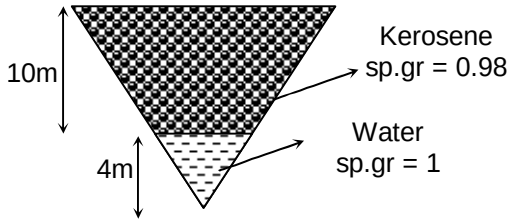
(C) $v=\sqrt{\frac{3}{2}g\ell\sin\theta}$ (D) $v=\sqrt{\frac{1}{2}g\ell\sin\theta}$

10. The displacement of the particle varies with time $x=24\sin\omega t-32\sin^3\omega t$. If its motion is SHM, then maximum acceleration is :

(A) $12\omega^2$ (B) $72\omega^2$

(C) $144\omega^2$ (D) $\sqrt{192}\omega^2$

11. The figure shows a conical container of half-apex angle 37° filled with certain quantities of kerosene and water. The force exerted by the water on the kerosene is approximately. (Take atmospheric pressure = 10^5Pa, g = 10 m/s^2) (container is open at the top)



(A) 3×10^7 N
(B) 4×10^7 N
(C) 5×10^7 N
(D) 5.6×10^6 N

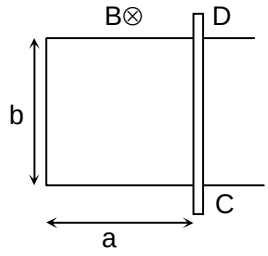
12. A calorimeter (of water equivalent 50g) contains 250g of water and 50g of ice at 0°C, 30g of water at 80°C is added to it. The final condition of the system will be:
[s_{water} = 1cal/g-C, L_f = 80cal/g]
(A) temperature of the system will be 4.2°C
(B) temperature of the system will still be 0°C and the entire ice will melt
(C) the temperature will be 0°C and half of ice will melt
(D) the temperature will be 0°C and 20g of ice will be left

13. There are two thin spheres A and B of the same material and same thickness. They emit like black bodies. Radius of A is double that of B. A and B have same temperature T . When A and B are kept in a room of temperature T_0(<T) , the ratio of their rates of cooling (rate of fall of temperature) is : [assume negligible heat exchange between A and B]
(A) 2 : 1
(B) 1 : 1
(C) 4 : 1
(D) 8 : 1

14. A sphere of radius R is having uniform charge density ρ. The energy density of the electric field inside sphere, at a distance r(r<R) is given by (in J/m^3) : (take dielectric constant inside the sphere as 1)
(A) $U_E = \dfrac{\rho^2 r^2}{9\epsilon_0}$
(B) $U_E = \dfrac{\rho^2 r^2}{18\epsilon_0}$
(C) $U_E = \dfrac{\rho^2 r^2}{4\epsilon_0}$
(D) $U_E = \dfrac{\rho^2 r^2}{5\epsilon_0}$

15. A satellite is revolving round the earth in an orbit of radius r with time period T . If the satellite is revolving round the earth in an orbit of radius r + Δr (Δr<<r) with time period T + ΔT then :
(A) $\dfrac{\Delta T}{T} = \dfrac{3}{2}\dfrac{\Delta r}{r}$
(B) $\dfrac{\Delta T}{T} = \dfrac{2}{3}\dfrac{\Delta r}{r}$
(C) $\dfrac{\Delta T}{T} = \dfrac{\Delta r}{r}$
(D) $\dfrac{\Delta T}{T} = -\dfrac{\Delta r}{r}$

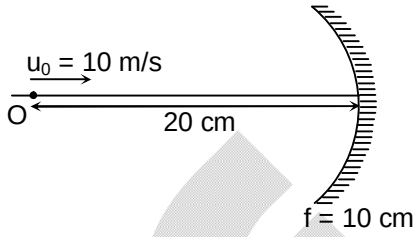
16. A U-shaped conducting frame is fixed in space. A conducting rod CD lies at rest on the smooth frame as shown. The frame is in uniform magnetic field B_0, which is perpendicular to the plane of frame. At time t = 0 , the magnitude of magnetic field begins to change with time t as, $B = \dfrac{B_0}{1+kt}$, where k is a positive constant. For no current to be ever induced in frame, the speed with which rod should be pulled starting from time t = 0 is (the rod CD should be moved such that its velocity must lie in the plane of frame and perpendicular to rod CD)



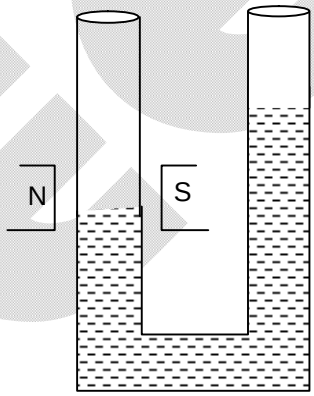
(A) ak
(B) bz
(C) a(1 + kt)
(D) b(1 + et)

17. In an electromagnetic wave, direction of propagation is in the direction of :
(A) $\vec{E}$ (B) $\vec{B}$
(C) $\vec{E}\times\vec{B}$ (D) None of these

18. An object moves with a uniform velocity u_0 = 10 m/s along the axis of a concave spherical mirror of focal length 10 cm. When the object is at the centre of curvature, the magnitude of acceleration of image at this instant is a m/s^2. Find a.
(A) 20
(B) 5
(C) 25
(D) None of these



19. In figure a U shaped tube contains magnetic material in which one arm of tube is kept between the poles of magnet. Which type of magnetic material is this
(A) Ferro magnetic
(B) Diamagnetic
(C) Paramagnetic
(D) None of these



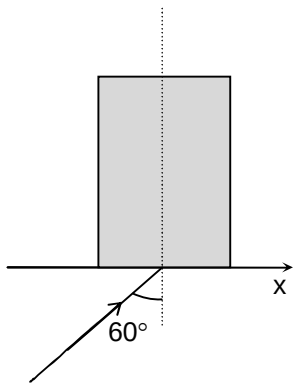
20. When Silicon p-n junction is in forward biased condition with series resistance, it has knee voltage 0.5 Volt. Current flows in it is 5 mA, when p-n junction is connected with 2.5 volt battery, the value of series resistance is
(A) 200 Ω (B) 100 Ω
(C) 500 Ω (D) 400 Ω

SECTION – B

(Numerical Answer Type)

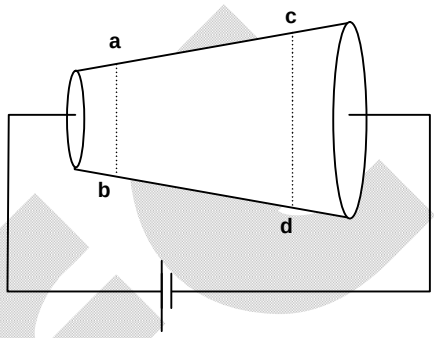
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A ray of light enters into a thick glass slab from air as shown in the figure. The refractive index varies as $\mu = (3\sqrt{3} - y)$. If width of slab is very large then maximum value of y-coordinate of the ray is $\frac{N\sqrt{3}}{2}$. Find the value of N.

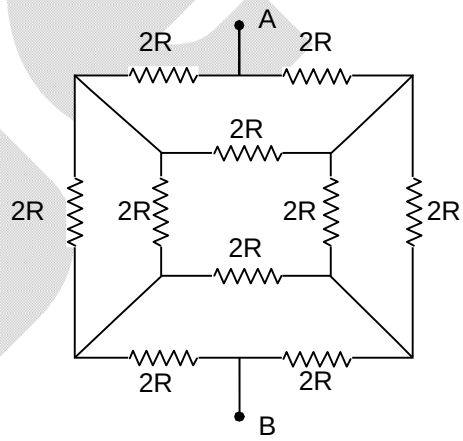


22. The densities of two cylinders A and B of same radii R and same length vary with radial distance as $\rho_{A(r)} = k\left(\frac{r}{R}\right)^2$ and $\rho_{B(r)} = k\left(\frac{r}{R}\right)^3$ respectively, where k is a constant. The moment of inertia of the individual cylinders about passing through their centers are I_A and I_B respectively, then the ratio of $\frac{I_A}{I_B}$ is $\frac{7}{N}$. Find the value of N.

23. A metallic wire has variable cross sectional area. Cross sectional area at cd is four times the area at ab. The wire is connected to a cell as shown in the figure. The ratio of power dissipated per unit volume at section cd to that at section ab is 4N : 1. Find the value of N.

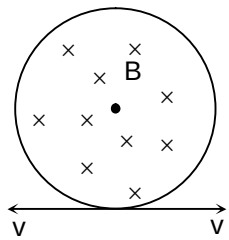


24. In the circuit shown in figure. The equivalent resistance across AB is $\frac{NR}{2}$. Find the value of N.



25. A particle having mass m and charge q is projected with a velocity v making an angle $\theta = 30°$ with the direction of a uniform magnetic field B. The magnitude of change in velocity of particle after time $t = \frac{3}{2}\frac{\pi m}{qB}$ is $\frac{v}{\sqrt{N}}$. Find N.

26. A circular conducting loop of radius r_0 and having resistance per unit length λ as shown in the figure is placed in a magnetic field B which is constant in space and time. The ends of the loop are crossed and pulled in opposite directions with a velocity v such that the loop always remains circular and the radius of the loop goes on decreasing then find the current induced in the loop in ampere. (take B = 4 Tesla, $\lambda = 2\Omega/m$ and $v = \pi$ m/s)

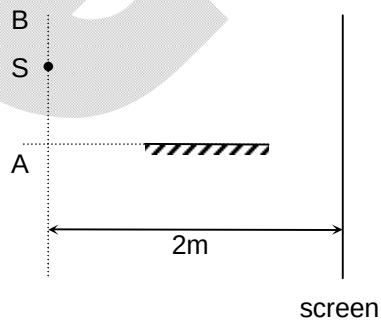


27. A parallel plate capacitor has square plates of side length L kept at a separation d. The space between them is filled with a dielectric constant which changes on $k = \frac{B}{x^2}$, where B is positive constant and x is the distance measurred from left plate towards right plate. Equivalent capacitacne is $\frac{NBL^2\varepsilon_0}{d^3}$. Find value of N.

28. In a hypothetical system a particle of mass m and charge $-3q$ is moving around a very heavy particle having charge q. Assuming Bohr's model to be true to this system. The orbital velcoity of mass m when it is nearest to heavy particle is $v = \frac{Nq^2}{2h\varepsilon_0 n}$. Find the value of N. (h = Plank's constant and n = principal quantum number and ε_0 = permitivity) (neglect gravitational force between the two particles)

29. In a stationary wave that forms as a result of reflection of waves form an obstacle, the ratio of the amplitude at an antinode to the amplitude at a node is 4. The percentage of energy transmitted is $8 \times$ N. Find the value of N.

30. In an interference experiment (the arragnement is shown in the figure) the source and screen are seperated by a distance 2m. At a certain position of source the fringe width is $\frac{1}{4}$mm and by moving along line AB by 0.6 mm, the fringe width changed to $\frac{1}{6}$mm. The wavelength of light used is $N \times 10^{-7}$m. Find the value of N.



Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Given two standard reduction potential below, what is the K_{SP} of Ag_2SO_4 at 25°C? [Given antilog of (0.0598 = 1.1476]

$Ag_2CrO_4(s) + 2e^- \longrightarrow 2Ag(s) + CrO_4^{2-}(aq)$ $\quad E^o = +0.446\,V$

$Ag^+(aq) + e^- \longrightarrow Ag(s)$ $\quad E^o = +0.799\,V$

(A) 8.64×10^{11} (B) 1.08×10^{-6}

(C) 1.15×10^{-12} (D) 1.11×10^{-39}

32. A 0.10 M solution of glycolic acid is 3.8% ionized. What is the % ionization of 0.50 M solution of glycolic acid?

(A) 1.74% (B) 3.8%

(C) 8.5% (D) 19.0%

33. A 10.0 gm sample of a soluble barium salt is treated with an excess of sodium sulphate to precipitate 8.93 gm $BaSO_4$ (M = 233.4). Which is the barium salt is it?

(A) $BaCl_2\,(M = 208.2)$ (B) $Ba(O_2CH)_2\,(M = 227.3)$

(C) $Ba(NO_3)_2\,(M = 261.3)$ (D) $BaBr_2\,(M = 297.1)$

34. Which compound when dissolve in water will increase the electrical conductivity of water least?

(A) SO_3 (B) H_2O_2

(C) $MgCl_2$ (D) NH_4Br

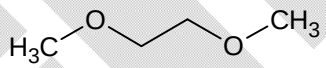
35. Which metal react most vigorously with water?

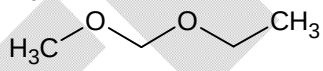
(A) Aluminimum (Al) (B) Zinc (Zn)

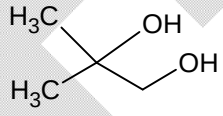
(C) Cesium (Cs) (D) Mercury (Hg)

36. Which isomer of $C_4H_{10}O_2$ has highest normal boiling point?

(A) 

(B) 

(C) 

(D) 

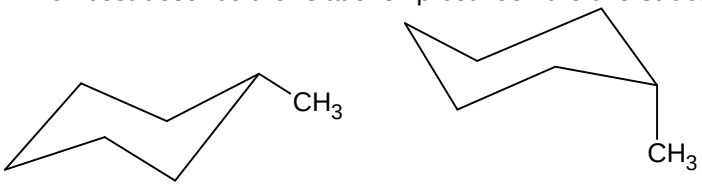
37. Under certain conditions, the rate of decomposition of hydrogen peroxide is 1st order in H_2O_2. It is observed that the concentration of H_2O_2 decreases by1.00% in 300 sec. What is the rate constant for the reaction?

(A) $3.35 \times 10^{-5}\ S^{-1}$ (B) $2.31 \times 10^{-3}\ S^{-1}$

(C) $3.33 \times 10^{-3}\ S^{-1}$ (D) $300\ S^{-1}$

38. Which of the following is correct regarding the IO_3^- ion?
(A) Triangular planar with 6 lone pair
(B) Triagonal pyramidal with 7 lone pair
(C) Triangular planar with 7 lone pair
(D) Tetrahedral with 6 lone pair

39. Which best describe the relationship between the two structure shown below:

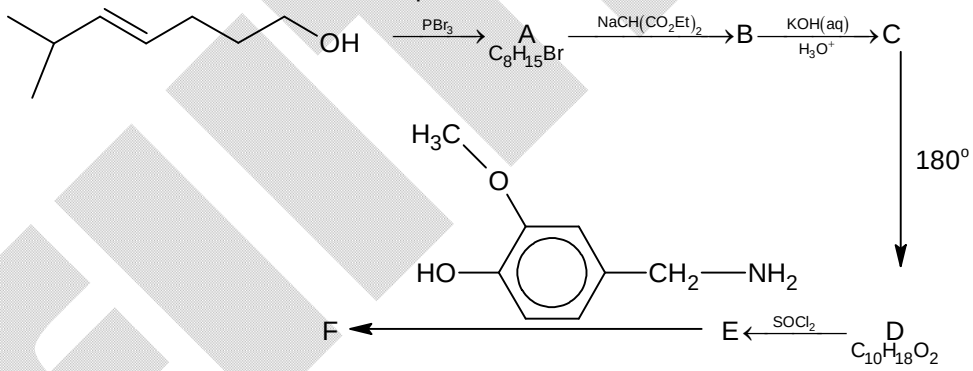


(A) They are identical
(B) They are enantiomers
(C) They are in rapid equilibrium at room temperature and are present in equal quantities
(D) They are in rapid equilibrium at room temperature and are present in unequal quantities

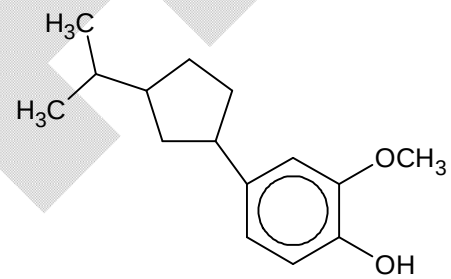
40. In the reaction of sodium cyanide with 1-bromo butane in methanol solution. What is the principal role of the cyanide ion?
(A) A Bronsted base
(B) A nucleophile
(C) An oxidant
(D) A solvent

41. Element A has 3-valence electron and element B has 6-valence electron. Element A and B are in same period of the periodic table. What is the likely formula of the compound that element A and B form together.
(A) AB
(B) AB_2
(C) A_2B_3
(D) A_3B_3

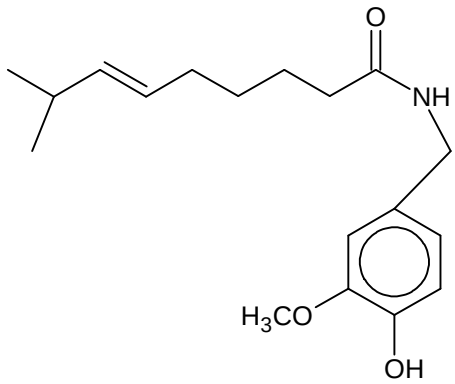
42. The test of pepper is mainly due to capsaicinoids. The synthetic route of capasaicin F is shown below. The structural formula of capasaicin F is:



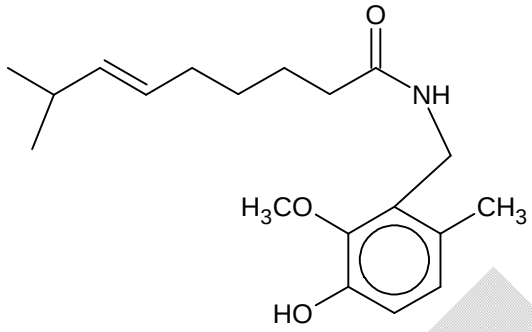
(A)



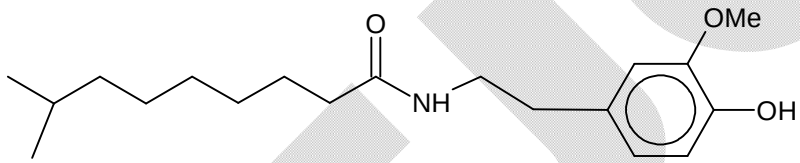
(B)



(C)



(D)



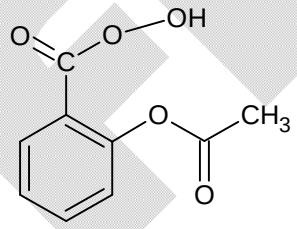
43. For how many seconds must a current of 5.00 A flow in order to deposit 1.30 gm of Nickel from solution of Nickel (II) nitrate? (Atomic mass of Ni = 58.69 gm)

(A) 2.14×10^2 S (B) 4.28×10^2 S

(C) 8.55×10^2 S (D) 4.28×10^3 S

44. Aspirin has the structural formula given below. Which functional groups does this molecule contain?

I. Acid
II. Alcohol
III. Ester
IV. Ether



(A) I and III (B) II and III

(C) II and IV (D) I, II and III

45. In which species does sulphur have the lowest oxidation state?

(A) SCl_2 (B) OSF_2

(C) H_2SO_3 (D) SF_6

46. How many H^+ ions are required when the equation below is balanced with the smallest whole number of coefficients?

$$Cu(s) + NO_3^-(aq) + H^+(aq) \longrightarrow Cu^{2+}(aq) + NO(g) + H_2O(\ell)$$

(A) 2 (B) 4
(C) 6 (D) 8

47. Which species contain only covalent bonds?

(A) AlF_3 (B) NH_4NO_3
(C) H_2SO_4 (D) $K_2Cr_2O_7$

48. What would be the bond length of C = O bond in $\left[Fe(CO)_5\right]$ if isolated C = O bond length is $1.128\ \overset{o}{A}$

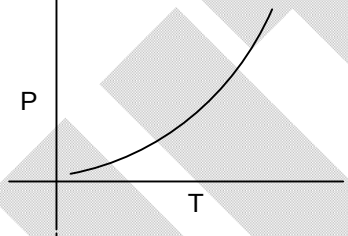
(A) $1.128\ \overset{o}{A}$ (B) $1.115\ \overset{o}{A}$
(C) $1.72\ \overset{o}{A}$ (D) $1.15\ \overset{o}{A}$

49. If λ_o is the threshold wave length of incident light and λ is wavelength of incident light. The velocity of photoelectron ejected from the metal surface is

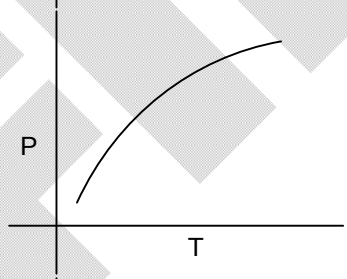
(A) $\sqrt{\dfrac{2h(\lambda_o - \lambda)}{m}}$ (B) $\sqrt{\dfrac{2hc(\lambda_o - \lambda)}{m}}$
(C) $\sqrt{\dfrac{2hc(\lambda_o - \lambda)}{m(\lambda.\lambda_o)}}$ (D) $\sqrt{\dfrac{2hc}{m}\left(\dfrac{1}{\lambda_o} - \dfrac{1}{\lambda_o}\right)}$

50. Which graph best represent the vapour pressure of water as a function of temperature from 0^oC to 100^oC?

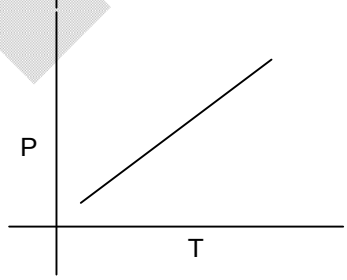
(A)



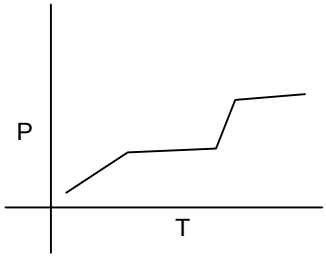
(B)



(C)



(D)



SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The value of the quantum number ℓ for a 5p orbital is ……..

52. The total number of sum of σ and π bond present in mesitylene used as a precursor is……..

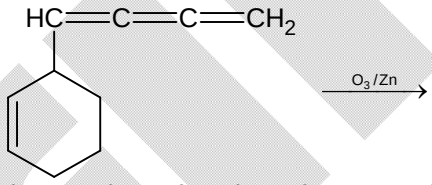
53. How many anion will give colourless acid vapour/gas with concentration H_2SO_4 on reaction with following anions.
$CH_3COO^-, Cl^-, Br^-, S^{2-}, SO_3^{2-}, BO_3^{3-}, NO_3^-, C_2O_4^{2-}, I^-$

54. How many π electrons are present in ferrocene?

55. A decapeptide (M.wt. = 796) on complete hydrolysis give gly(mol. Wt. 75), ala-phe-gly (mol. Wt. 75), ala-phe-gly contributes 47% of the wt. of hydroysed product. The number of glycine unit present in decapeptide is……

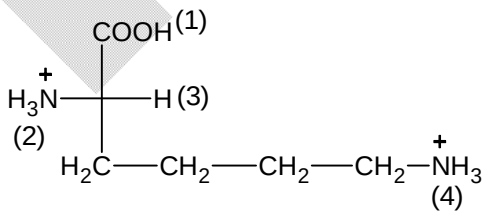
56. The resistance of 0.5 M solution of an electrolyte in a cell was found to be 50Ω. If the electrode in the cell are 2.2 cm apart and have an area of 4.4 cm^2 then the molar conductivity of the solution is $x \times 10^{-3}$ S m^2 mol^{-1}. Then the value of x is……

57. Compound B $\xrightarrow{O_3/Zn/H_2O}$ $C_3H_2O_3$ (only)
Then the value of double bond equivalent of B is

58. 

The number of moles of CO_2 evolved in the oxidation ozonolysis of the above compound is……

59. In strongly acidic solution, L-lysine is primarily a dication as shown below. As the pH is raised, which proton is lost to form monocation?



60. In 2D sheet silicate the number of oxygen atoms share per tetrahedron unit is…….

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. In ΔABC, we have $\overrightarrow{AB}\cdot\overrightarrow{AC}+2\overrightarrow{BA}\cdot\overrightarrow{BC}=3\overrightarrow{CA}\cdot\overrightarrow{CB}$, then the maximum value of sin C is

(A) $\frac{\sqrt{7}}{3}$ (B) $\frac{\sqrt{7}}{4}$

(C) $\frac{\sqrt{5}}{3}$ (D) $\frac{\sqrt{11}}{4}$

62. Given that f(x) is an odd function on R. $f(1)=1$, and $f\left(\frac{x}{x-1}\right)$ = x f(x) for any x < 0, then the value of

$f(1)f\left(\frac{1}{100}\right)+f\left(\frac{1}{2}\right)f\left(\frac{1}{99}\right)+f\left(\frac{1}{3}\right)f\left(\frac{1}{98}\right)+\ldots+f\left(\frac{1}{50}\right)f\left(\frac{1}{51}\right)$ is

(A) $\frac{2^{100}}{99!}$ (B) $\frac{2^{99}}{100!}$

(C) $\frac{2^{99}}{99!}$ (D) $\frac{2^{98}}{99!}$

63. Let f(x) be defined on R. satisfying for each real number x, f(x + 3) f(x – 4) = – 1. More over, when $0\le x<7, f(x)=\log_2(9-x)$. Then f(–100) equals

(A) $-\frac{1}{2}$ (B) -1

(C) -2 (D) $-\frac{1}{4}$

64. Let x, y be real numbers such that x^2 + 2cosy = 1. The range of x – cos y is

(A) $[-1,\sqrt{3}-1]$ (B) $[-1,\sqrt{3}+1]$

(C) $[1,\sqrt{3}+1]$ (D) $[\sqrt{3}-1,1]$

65. A three-digit positive integer is called a “stable number” if any two consecutive digits differ by at most 1. The number of stable numbers is

(A) 95 (B) 70

(C) 63 (D) 75

66. Suppose complex numbers z, w satisfy |z| = 3 and $(z+\bar{w})(\bar{z}-w)=7+4i$, where i is the imaginary unity, and $\bar{z},\bar{w}$ are conjugates of z, w, respectively. Then $(z+2\bar{w})(\bar{z}-2w)$ is

(A) $-1+8i$ (B) $1+8i$

(C) $1-8i$ (D) $-1-8i$

67. None of three positive real numbers u, v, w being 1, if $\log_u vw + \log_v w = 5$ and $\log_v u + \log_w v = 3$, then the value of $\log_w u$ is

(A) $\frac{5}{4}$ (B) $\frac{4}{5}$

(C) $\frac{3}{4}$ (D) $\frac{4}{3}$

68. Given that the quadratic equation $ax^2 + bx + c = 0$ has no real roots, but Adam got two roots 2 and 4 since he wrote down a wrong value of a. Ben also got two roots – 1 and 4 because he wrote down the sign of a term wrongly, then the value of $\frac{2b+3c}{a}$

(A) 3 (B) 4

(C) 5 (D) 6

69. The units digits of $\left[\frac{10^{20000}}{10^{100}+3}\right]$ is (where [.] denotes G.I.F)

(A) 1 (B) 2

(C) 3 (D) 4

70. Let $x_1, x_2, \ldots, x_n$ be a sequence of integers such that

(i) $-1 \le x_i \le 2$, for $i = 1,2,\ldots,n$;

(ii) $x_1 + x_2 + \ldots + x_n = 19$;

(iii) $x_1^2 + x_2^2 + \ldots + x_n^2 = 99$.

The minimum and maximum possible values of $x_1^3 + x_2^3 + \ldots + x_n^3$ are m and M respectively, then M – m is

(A) 114 (B) 90

(C) 118 (D) 80

71. Let f(x) be and odd function on R. and $f(x) = x^2$ for $x \ge 0$. Suppose for any $x \in [a, a+2], f(x+a) \ge 2f(x)$. Then the range of real number a is

(A) $\left[-\sqrt{2},\infty\right)$ (B) $\left[\sqrt{2},\infty\right)$

(C) $\left[-1,\infty\right)$ (D) $\left[1,\infty\right)$

72. The sum of all the positive integers n satisfying $\frac{1}{4} < \sin\frac{\pi}{n} < \frac{1}{3}$ is

(A) 30 (B) 36

(C) 33 (D) 39

73. $\lim\limits_{x\to 1} \frac{\sqrt[3]{x^2+7} - \sqrt{x+3}}{x^2 - 3x + 2}$

(A) $\frac{1}{12}$ (B) $\frac{1}{10}$

(C) $\frac{1}{20}$ (D) $\frac{1}{18}$

74. Let A be a real square matrix of order two such that $A^T - I = A^TA$, then which of the following is not correct ?

(A) A is symmetric matrix (B) $I + A^2 = 2A + A^4$

(C) $A^{-1} = I - A$ (D) $\det(A - I) = -1$

(where tr(X) denotes trace of a matrix X, X^T denotes transpose of X and I denotes unit matrix)

75. $\int \frac{3+2\cos x}{(2+3\cos x)^2}dx$ is equal to

(A) $\left(\frac{\sin x}{3\cos x+2}\right)+c$ (B) $\left(\frac{2\sin x}{3\sin x+2}\right)+c$

(C) $\left(\frac{2\cos x}{3\cos x+2}\right)+c$ (D) $\left(\frac{2\sin x}{3\cos x+2}\right)+c$

76. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by (a, b) R (c, d) if ad (b + c) = bc (a + d), then R is

(A) symmetric only (B) reflexive only

(C) transitive only (D) an equivalence relation

77. Suppose, a population A has 100 observations 101, 102, …… , 200 and another population B has 100 observations 151, 152, …… , 250. If V_A and V_B represent the variances of the two population respectively, then $\frac{V_A}{V_B}$ is

(A) $\frac{9}{4}$ (B) $\frac{4}{9}$

(C) $\frac{2}{3}$ (D) 1

78. If $\alpha = \frac{1}{3}\sin^{-1}\left(\frac{2x}{1+x^2}\right)+\frac{1}{3}\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ where $x \geq \frac{4}{3}$, then the value of $\frac{\cos 2\alpha + \sec\alpha + 3\sqrt{3}}{\sqrt{3}}$ is equal to

(A) 3 (B) $2+\sqrt{3}$

(C) $\frac{3\left(\sqrt{3}+1\right)}{\sqrt{3}}$ (D) $\left(\frac{\sqrt{3}}{2}+3\right)$

79. If $\int_0^{\pi}\left(\frac{x}{1+\sin x}\right)^2 dx = A$, then the value for $\int_0^{\pi}\frac{2x^2\cdot\cos^2 x/2}{(1+\sin x)^2}dx$ is equal to

(A) $A + 2\pi - \pi^2$ (B) $A - 2\pi - \pi^2$

(C) $2\pi - A - \pi^2$ (D) None of these

80. A curve y = f(x) passes through the origin. Through any point (x, y) on the curve, lines are drawn parallel to the coordinate axes. If the curve divides the area formed by these lines and coordinate axes in the ratio m : n. Then the equation of curve is

(A) $y = Cx^{m/n}$ (B) $my^2 = Cx^{m/n}$

(C) $y^3 = Cx^{m/n}$ (D) none of these

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Given $f(x)=\left(\sin\frac{kx}{10}\right)^4+\left(\cos\frac{kx}{10}\right)^4$, where k is a positive integer. If for any real number a, $\{f(x)|a<x<a+1\}=\{f(x)|x\in R\}$ then the minimum of k is _____.

82. Let hyperbola $C: x^2-\frac{y^2}{3}=1$, with the left and right focal points being F_1, and F_2 respectively. Crossing F_2 draw a line that intersects with the right half of the hyperbola at points P and Q, making $\angle F_1PQ = 90^o$. The inscribed circle radius of ΔF_1PQ is $\sqrt{k}-1$, then the value of k is

83. In the Cartesian plane, ellipse C is given by $\frac{x^2}{9}+\frac{y^2}{10}=1$, point F is the upper focus of C, A is the right vertex of C, O is the origin and P is a moving point on C in the first quadrant. The maximum area of quadrilateral OAPF is $\frac{3\sqrt{k}}{2}$, then the value of k is

84. A plane point set is given by K = {(x, y) | x, y = – 1, 0, 1}. Choose 3 points from K at random. Then the probability that there are two points among them with distance $\sqrt{5}$ is $\frac{4}{\alpha}$, then the value of α is

85. Let f(x) be defined on R, such that $f(x) + x^2$ is an odd function, and $f(x) + 2^x$ is an even function. Then $4|f(1)|$ equals

86. If one side of square ABCD is on the line y = 2x – 17, and the other two vertices lie on parabola $y = x^2$. Then the minimum area of the square is _____

87. We arrange the expansion of $\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n$ in decreasing powers of x. If the coefficients of first three terms of form an arithmetic progression, then, in the expansion, there are _____ terms of x with integer power.

88. The number of points of discontinuity of the function $f:(-2,2)\to R$, $f(x)=\left[2^{-x^2}\left[2^{x^2}\right]\right]$ (where $[.]$ represent G.I.F.) is

89. The minimum of $y=\left(a\cos^2 x-3\right)\sin x$ is – 3, then the maximum value of real number a is_____

90. Let $C_1 \equiv (x-1)^2 + y^2 = 1$ and $C_2: x^2 + y^2 = r^2$ intersect at P in 1st quadrant and let R be the point where the line passing through the points Q ≡ (0, r) and P intersects the x-axis. If radius of C_2 decreases indefinitely then limiting position of R is given by (α, 0) then 'α' is equal to